

the constraint set

$Sv = 0$: exact when the alternation ends on the balance, else a residual
$\ell \le v \le u$: exact when it ends on the box, else a penalty
second law $v_j \Delta_r G_j \le 0$: a hinge, or with tdg a directional box
protein pool $\sum_g \mathrm{MW}_g E_g \le p_{\mathrm{avail}}$: a hinge
Gibbs dissipation: a logged diagnostic; the anchored registry entry carried it as a hinge at nonzero constraint scale

measured (rounds 2 to 10)

never reaches the polytope: 58 percent of reactions out of the box at the balance end, balance residual 0.85 to 2.0 at the box end; the growth column is a free signed coordinate